Slide-by-Slide Lecture Plan

ENG-654 — Kinematics-Grounded Motion Planning for Robots

Instructor planning document

Abstract

This document specifies the content, timing, teaching purpose, and recommended visual material for each of the eight 75-minute lectures in the four-session course. Image notes describe what should be drawn, animated, or captured from simulation; they are not generic decoration.

Contents

Course rhythm

Each four-hour session follows the same schedule: 08:15–09:30 lecture, 09:30–09:45 break, 09:45–11:00 lecture, 11:00–11:15 break, and 11:15–12:00 guided exercise. The eight lectures follow the progression

read the robot → solve the robot → understand its solution structure → plan feasible motion.

The timings below sum to 75 minutes per lecture. When a derivation is done live, reduce the number of displayed algebraic steps accordingly.

1 Session 1, Lecture 1: Introduction, Frames, Transformations, and DH Modelling

Central question

How do we convert a physical robot into a reproducible mathematical model?

Slide 1 — Opening motivation: reachable is not always executable — 5 min

On-slide content. Show the course title and three claims: a reachable pose may not admit a continuous path; one pose may have several joint solutions; and every executed path must respect the robot's geometry.

Teaching note. Begin with a concrete manipulation failure rather than definitions. Describe a robot that can touch both endpoints of a Cartesian line but cannot track the complete line because of a joint limit or singularity.

Image/figure note. Use a full-width robot image with the desired Cartesian curve drawn in blue, a blocked posture in red, and the start and goal poses in green.

Slide 2 — The four-session narrative — 5 min

On-slide content. Present the sequence: read the robot, solve the robot, understand its singular and branch structure, and plan motion using that structure.

Teaching note. Explain that the same representative robot will reappear in all four sessions. This continuity reduces notation changes and lets students see why modelling choices matter later.

Image/figure note. Draw a horizontal flow diagram: physical robot → DH/PoE/URDF → IK solutions → singularity map → feasible trajectory.

Slide 3 — What information must be read from a robot? — 5 min

On-slide content. List links, joints, joint types, joint axes, geometric dimensions, coordinate frames, joint limits, tool frame, and collision geometry.

Teaching note. Distinguish the physical object from its model. A photograph may reveal the architecture, but not necessarily zero offsets, software frame conventions, or permitted joint ranges.

Image/figure note. Annotate a CAD model of a 6R arm with labels for the base, links, revolute axes, wrist center, tool frame, and one joint limit interval.

Slide 4 — Links, joints, and serial chains — 5 min

On-slide content. Define rigid link, revolute joint, prismatic joint, joint coordinate, serial chain, and end-effector.

Teaching note. Clarify that the joint is a relative-motion constraint between two links, while q_i is the scalar coordinate used to describe its allowed motion.

Image/figure note. Use a clean 3R mechanism drawing with each link numbered and each joint shown using a conventional revolute-joint symbol.

Slide 5 — Configuration and configuration space 5 min

On-slide content. Define $\mathbf{q} = [q_1, \ldots, q_n]^T$ and explain configuration space as the set of all admissible joint vectors.

Teaching note. Use a 2R robot to distinguish one configuration from the entire configuration space. Mention periodic revolute coordinates and the restriction introduced by mechanical joint limits.

Image/figure note. Place a 2R arm at one pose beside a (q_1, q_2) chart. Optionally show the ideal torus and then a bounded rectangle representing joint limits.

Slide 6 — Task space and workspace are different from joint space 5 min

On-slide content. Define task variables, end-effector pose, and workspace. State that several configurations can map to one task-space pose.

Teaching note. Emphasize that a task may be specified in workspace, but physical continuity and collision feasibility are ultimately checked in configuration space.

Image/figure note. Draw a many-to-one mapping from several robot postures on the left to one end-effector frame on the right.

Slide 7 — Coordinate frames are part of the model 5 min

On-slide content. Introduce world, base, link, joint, and tool frames. State that coordinates are incomplete unless the reference frame is known.

Teaching note. Demonstrate that the same physical point has different numerical coordinates in two frames. Introduce the notation ${}^{A}\mathbf{p}_P$ and read it aloud.

Image/figure note. Show two frames attached to the same rigid body and one point P, with coordinate vectors expressed from each origin.

Slide 8 — Rotation matrices 5 min

On-slide content. Define ${}^{A}R_B$ and explain that its columns are the axes of frame B expressed in frame A.

Teaching note. Focus on interpretation rather than proof. Mention orthogonality and determinant $+1$, then ask students which direction each column represents.

Image/figure note. Color the three axes of frame B and use the same colors for the three columns of ${}^{A}R_B$.

Slide 9 — Translation vectors and homogeneous transformations 5 min

On-slide content. Present

$${}^{A}T_B = \begin{bmatrix} {}^{A}R_B & {}^{A}\mathbf{p}_B \\ \mathbf{0}^T & 1 \end{bmatrix}.$$

Define each block explicitly.

Teaching note. Explain that the translation is the position of the origin of B expressed in A. Show how the matrix acts on a point in homogeneous coordinates.

Image/figure note. Use a block-annotated matrix beside frames A and B, with arrows linking the rotation and translation blocks to the geometry.

Slide 10 — Composition and order of transforms 5 min

On-slide content. State ${}^{A}T_C = {}^{A}T_B{}^{B}T_C$ and explain frame-index compatibility.

Teaching note. Read the equation as a path through frames. Warn that rigid transformations do not generally commute, so changing order changes the physical motion.

Image/figure note. Draw frames A, B, and C connected by arrows. Highlight the repeated B index as the compatible intermediate frame.

Slide 11 — Forward kinematics of a serial robot 5 min

On-slide content. Present ${}^{0}T_n = {}^{0}T_1{}^{1}T_2 \cdots {}^{n-1}T_n$ and identify the position and orientation contained in the final matrix.

Teaching note. Connect frame composition to the robot chain. State the forward-kinematics question: given $\mathbf{q}$, where is the tool?

Image/figure note. Place coordinate frames on every link of a serial arm and print the corresponding transform product underneath.

Slide 12 — Why use a systematic frame convention? 5 min

On-slide content. Explain that many frame assignments can describe the same robot, but a convention makes the parameterization reproducible and tabulatable.

Teaching note. Motivate DH as an organizational tool rather than an arbitrary ritual. Show how inconsistent frame choices make model comparison difficult.

Image/figure note. Show the same 3R robot with two different valid frame assignments, then a third standardized DH assignment.

Slide 13 — The four Denavit–Hartenberg parameters 5 min

On-slide content. Define a_i, α_i, d_i, and θ_i geometrically; identify which parameter varies for revolute and prismatic joints.

Teaching note. Spend time on the common normal and the sign of the twist angle. Avoid presenting only the expanded matrix before students understand the geometry.

Image/figure note. Draw two consecutive joint axes and their common normal. Label all four parameters directly on the drawing.

Slide 14 — DH frame assignment and worked table 5 min

On-slide content. Give the procedure: align z with joint axes, choose x along a common normal, place origins, and complete right-handed frames. Fill a three-row table for the running 3R robot.

Teaching note. Reveal one row at a time and ask students to predict the next parameter. Discuss intersecting and parallel-axis special cases.

Image/figure note. Use a split slide: large robot drawing on the left and a partially completed DH table on the right.

Slide 15 — From DH table to validation 5 min

On-slide content. Show ${}^{i-1}T_i = R_z(\theta_i)T_z(d_i)T_x(a_i)R_x(\alpha_i)$, form the full product, and list common mistakes: wrong axis, mixed convention, sign error, and incorrect zero offset.

Teaching note. Finish by stating that a model is not trustworthy until it is checked numerically or in simulation at several configurations. Preview PoE and URDF as alternative descriptions of the same physical robot.

Image/figure note. Draw a pipeline: DH table → link transforms → forward pose → simulation overlay and error check.

2 Session 1, Lecture 2: Product of Exponentials and Reading URDF Models

Central question

How can the same robot be represented through joint screw axes and a computational kinematic tree?

Slide 1 — The same robot in three languages 5 min

On-slide content. Place the running robot beside a DH table, a set of screw axes, and a URDF tree. State that all three descriptions must predict the same end-effector pose.

Teaching note. Explain that the goal is not to choose a winner. DH, PoE, and URDF emphasize different information and are useful in different workflows.

Image/figure note. Use a three-column comparison with the same robot centered above the three representations.

Slide 2 — Why introduce Product of Exponentials? 5 min

On-slide content. Contrast consecutive-frame parameters with direct descriptions of physical joint axes. State that PoE expresses forward kinematics as a product of rigid motions generated by joints.

Teaching note. Point out that PoE avoids assigning a DH frame to every link, but still requires a reference frame, a zero configuration, and a consistent twist convention.

Image/figure note. Draw a DH chain of frames on the left and fixed screw axes in the home configuration on the right.

Slide 3 — Motion generated by a revolute joint 5 min

On-slide content. For a unit axis $\boldsymbol{\omega}$ through point $\mathbf{r}$, show that a point velocity is $\mathbf{v}_P = \boldsymbol{\omega} \times (\mathbf{p} - \mathbf{r})$.

Teaching note. Use the geometry to explain angular and tangential velocity. This prepares the linear part of a revolute twist.

Image/figure note. Draw the axis, point $\mathbf{r}$, point P, angular-velocity arrow, radius vector, and tangential velocity arrow.

Slide 4 — Twist coordinates of revolute and prismatic joints 5 min

On-slide content. Define the ordering convention $\mathcal{S} = (\boldsymbol{\omega}, \mathbf{v})$, with $\mathbf{v} = -\boldsymbol{\omega} \times \mathbf{r}$ for a revolute joint and $\mathcal{S} = (\mathbf{0}, \mathbf{v})$ for a prismatic joint.

Teaching note. State the ordering convention prominently and keep it unchanged in all later Jacobian slides. Explain that two coordinate conventions in the literature are common.

Image/figure note. Show one revolute and one prismatic joint, each color-matched to its six-vector coordinates.

Slide 5 — Matrix representation of a twist 5 min

On-slide content. Introduce

$$[\mathcal{S}] = \begin{bmatrix} [\boldsymbol{\omega}] & \mathbf{v} \\ \mathbf{0}^T & 0 \end{bmatrix}$$

and define the skew-symmetric matrix $[\boldsymbol{\omega}]$.

Teaching note. Explain that this matrix belongs to the tangent representation of rigid motion and is the object exponentiated to obtain a finite transformation.

Image/figure note. Use arrows from the angular and linear components into the corresponding blocks of the matrix.

Slide 6 — Exponential motion of one joint 5 min

On-slide content. Interpret $e^{[\mathcal{S}]q}$ as the rigid transformation produced by moving one joint by amount q.

Teaching note. For a revolute joint, emphasize rotation about an arbitrary spatial axis. For a prismatic joint, emphasize translation along a fixed direction. Keep Rodrigues' formula in backup material.

Image/figure note. Show before-and-after poses of one rigid body for a revolute axis and for a prismatic axis.

Slide 7 — Home configuration and space screw axes 5 min

On-slide content. Define $M = T(\mathbf{0})$ and explain that each space screw axis $\mathcal{S}_i$ is expressed in the base frame at the chosen zero configuration.

Teaching note. Clarify that the zero configuration is a modelling choice, not necessarily a central or comfortable robot posture.

Image/figure note. Show the robot at $\mathbf{q} = \mathbf{0}$ with all joint axes drawn and the tool frame used to define M.

Slide 8 — Space-frame Product of Exponentials 5 min

On-slide content. Present

$$T(\mathbf{q}) = e^{[\mathcal{S}_1]q_1} e^{[\mathcal{S}_2]q_2} \cdots e^{[\mathcal{S}_n]q_n} M.$$

Teaching note. Read the expression carefully and explain that the exponential factors are ordered according to the serial chain. Relate each factor to one physical joint motion.

Image/figure note. Draw the serial robot with numbered axes and place the matching exponential factors beneath them.

Slide 9 — Worked PoE model of the running 3R robot 5 min

On-slide content. Derive $\mathcal{S}_1$, $\mathcal{S}_2$, $\mathcal{S}_3$, and M from the home configuration.

Teaching note. Fully derive at least one revolute twist from an axis direction and a point on the axis. Let students propose the remaining twists before revealing them.

Image/figure note. Use a large, uncluttered robot diagram with the reference points and axis directions required to compute each twist.

Slide 10 — Body-frame PoE as an alternative 5 min

On-slide content. Briefly introduce $T(\mathbf{q}) = Me^{[\mathcal{B}_1]q_1} \cdots e^{[\mathcal{B}_n]q_n}$ and state that body screw axes are fixed in the end-effector frame at home.

Teaching note. Do not derive the adjoint relation in the main lecture unless time permits. The purpose is to prevent confusion when students encounter both formulas in references.

Image/figure note. Show space axes attached to the base and body axes attached to the tool in a side-by-side diagram.

Slide 11 — DH and PoE must agree 5 min

On-slide content. Compare $T_{DH}(\mathbf{q})$ and $T_{PoE}(\mathbf{q})$ for several test configurations and define position and orientation residuals.

Teaching note. Explain that symbolic expressions may look different while representing the same transformation. Numerical cross-validation is an essential modelling practice.

Image/figure note. Draw two modelling pipelines converging to the same pose, followed by residual plots near numerical zero.

Slide 12 — URDF as a rooted kinematic tree 5 min

On-slide content. Define URDF links and joints, parent-child relationships, and the rooted tree from base link to tool link.

Teaching note. State that URDF is not a DH table. A joint origin and axis describe a transform in a software frame convention, while visual and collision meshes are separate data.

Image/figure note. Show a robot model beside its link-joint tree with alternating link and joint nodes.

Slide 13 — Reading a URDF joint element 5 min

On-slide content. Display a short XML fragment containing joint type, parent, child, origin, axis, and limits. Explain each tag.

Teaching note. Highlight one line at a time and point to the corresponding element on the robot. Avoid showing a complete long URDF file.

Image/figure note. Place an annotated XML excerpt beside a simplified parent-link, joint-frame, and child-link drawing.

Slide 14 — URDF frame, axis, visual, and collision conventions 5 min

On-slide content. Explain that the joint origin places the joint frame relative to the parent link frame, the axis is expressed in the joint frame, and visual and collision geometry can use separate origins.

Teaching note. Emphasize that a robot may look visually correct while its collision geometry or kinematic axis is wrong. Mention inertial data only to distinguish it from kinematics.

Image/figure note. Show the same link as a detailed visual mesh, simplified collision mesh, and coordinate-frame skeleton.

Slide 15 — From URDF to a validated forward model 5 min

On-slide content. Give the procedure: traverse the tree from base to tool, compose fixed origin transforms and joint motion transforms, evaluate random configurations, and compare with DH and PoE.

Teaching note. Finish with the key distinction: DH organizes consecutive frames, PoE organizes joint motions, and URDF organizes a computational tree. Preview the exercise in which students reconstruct and validate two representations.

Image/figure note. Draw a complete validation loop: URDF parser → transform chain → simulator pose ↔ DH/PoE pose.

3 Session 2, Lecture 1: Analytical Inverse Kinematics of a Representative 3R Robot

Central question

How do we derive and enumerate every valid inverse-kinematic solution of a representative 3R robot?

Slide 1 — One target, several robot postures 5 min

On-slide content. Show one desired point and several 3R configurations that reach it. Ask whether an IK solver should return one configuration or the complete set.

Teaching note. Use the answers to motivate analytical IK. State that planning decisions are impossible to make globally when alternative branches are hidden.

Image/figure note. Superpose two or four robot postures using different line styles, while keeping the end-effector target fixed.

Slide 2 — Forward and inverse kinematics are different problems 5 min

On-slide content. Write $\mathbf{x} = f(\mathbf{q})$ for forward kinematics and $f(\mathbf{q}) = \mathbf{x}_d$ for inverse kinematics. List the possibilities: no solution, one solution, several isolated solutions, or a continuum.

Teaching note. Explain that forward kinematics is function evaluation, while inverse kinematics is equation solving. The inverse is generally not a single-valued function.

Image/figure note. Draw a many-to-one map from joint space to workspace, with several joint points mapping to one task-space point.

Slide 3 — Geometry and notation of the running 3R robot 5 min

On-slide content. Define joint variables, link dimensions, offsets, base frame, end-effector point, and the desired target coordinates.

Teaching note. State every geometric assumption. Use a robot that is general enough to exhibit multiple branches but simple enough for a complete derivation in one lecture.

Image/figure note. Use a large labelled mechanism drawing. Color q_1, q_2, and q_3 consistently with their appearances in later equations.

Slide 4 — Write the position equations 5 min

On-slide content. Present the three Cartesian position equations obtained from forward kinematics. Introduce shorthand such as $c_i = \cos q_i$ only after the full expressions are visible.

Teaching note. Link each term to a geometric contribution from a link. Avoid presenting an unexplained block of trigonometric expressions.

Image/figure note. Place the robot geometry beside the equations and color-match link contributions with equation terms.

Slide 5 — Use symmetry to isolate the base rotation — 5 min

On-slide content. Project the target into the horizontal plane and derive the equation for q_1 or for the radial distance seen by the remaining joints.

Teaching note. Explain why the first joint can be separated for the chosen architecture. Discuss shoulder alternatives if offsets produce more than one admissible azimuth solution.

Image/figure note. Show a top view with the target projection, base axis, azimuth angle, and radial distance.

Slide 6 — Reduce the remaining problem to a planar triangle — 5 min

On-slide content. Transform the target into the plane of joints 2 and 3 and define the reduced coordinates used in the arm-plane geometry.

Teaching note. This is the main conceptual reduction. Make clear which quantities are known after choosing a q_1 branch.

Image/figure note. Fade the spatial robot into a planar triangle containing joints 2 and 3 and the target point.

Slide 7 — Cosine-law equation for the elbow joint — 5 min

On-slide content. Derive an equation of the form $D = A + B\cos q_3 + C\sin q_3$, or the equivalent cosine-law expression for the selected geometry.

Teaching note. Explain the physical meaning of every side length and why the equation admits two regular solutions. Do not jump directly to a memorized formula.

Image/figure note. Draw the triangle with all side lengths, offsets, and included angles labelled.

Slide 8 — Solve the elbow branches — 5 min

On-slide content. Obtain the possible values of q_3 using a quadrant-safe expression. Identify elbow-up and elbow-down only as local geometric labels.

Teaching note. State the reachability condition explicitly, for example a cosine argument lying in $[-1, 1]$. Explain when two branches merge.

Image/figure note. Show the two arm-plane configurations reaching the same target and a small branch diagram for the two q_3 values.

Slide 9 — Recover the shoulder-plane angle — 5 min

On-slide content. Derive q_2 for each q_3 branch using angle decomposition or two atan2 terms.

Teaching note. Explain the geometry of the two component angles and why atan2 is safer than a single inverse cosine or inverse tangent.

Image/figure note. Label the line-of-sight angle and the internal triangle angle directly on the planar drawing.

Slide 10 — Complete the q_1 solutions — 5 min

On-slide content. Recover all admissible base-joint values and associate them with the corresponding (q_2, q_3) solutions.

Teaching note. Make the branch dependencies explicit. Some q_1 choices may require different transformed target coordinates, so avoid combining incompatible subsolutions.

Image/figure note. Use a branch tree whose first level is the base or shoulder choice and whose second level is the elbow choice.

Slide 11 — Organize the complete analytical solution set (5 min)

On-slide content. Present the candidate configurations as $\mathcal{Q}(\mathbf{x}_d) = \{\mathbf{q}^{(1)}, \ldots, \mathbf{q}^{(m)}\}$ and show how each branch was generated.

Teaching note. A branch tree is more informative than an unordered list. This representation will later become a layer in the analytical planning graph.

Image/figure note. Draw a clean branch tree ending in candidate joint vectors and branch labels.

Slide 12 — Reachability and workspace boundaries (5 min)

On-slide content. State the inequalities that classify a target as unreachable, on a boundary, or in a regular interior region.

Teaching note. Connect equality in the reachability condition to solution coalescence. Preview that these boundary configurations are often singular.

Image/figure note. Show a workspace cross-section divided into unreachable, boundary, and regular regions, with representative targets in each.

Slide 13 — Joint limits and angle wrapping (5 min)

On-slide content. Filter every analytical candidate using the robot's joint intervals. Explain equivalent representations such as q and $q + 2\pi$ before applying limits.

Teaching note. Distinguish a mathematical solution of the trigonometric equations from a valid configuration of the physical robot.

Image/figure note. Use a table containing candidate configuration, wrapped value, limit status, and final acceptance.

Slide 14 — Forward-kinematics verification (5 min)

On-slide content. For each accepted candidate, evaluate $f(\mathbf{q}^{(k)})$ and compute the target residual. State a numerical tolerance.

Teaching note. Emphasize that verification catches sign, branch-combination, and frame-convention errors. It should be automated in student code.

Image/figure note. Draw the validation pipeline and show the desired and reconstructed target points overlaid in a plot.

Slide 15 — Analytical and numerical IK: complementary tools (5 min)

On-slide content. Compare analytical enumeration with local iterative IK in terms of architecture dependence, completeness, initialization, runtime, and branch visibility.

Teaching note. Present a balanced view. Numerical IK is general and useful, but a single converged answer does not describe the complete solution structure. Preview the 6R and 7R extensions.

Image/figure note. Use a two-column comparison and a final graphic showing discrete 3R branches expanding into 6R branches and a 7R continuum.

4 Session 2, Lecture 2: Representative Wrist-Partitioned 6R and 7R Inverse Kinematics

Central question

How do wrist partitioning and continuous redundancy structure the inverse-kinematic solution set?

Slide 1 — From a target point to a target frame (5 min)

On-slide content. Contrast the 3R positioning problem with a 6R pose problem in which both the desired position $\mathbf{p}_d$ and orientation R_d must be matched.

Teaching note. Ask why the previous 3R solution is not sufficient. Explain that the tool frame, rather than only one point, is now prescribed.

Image/figure note. Show a 3R arm reaching a point beside a 6R arm aligning a complete coordinate frame.

Slide 2 — The wrist-partitioned 6R architecture (5 min)

On-slide content. Define a spherical wrist as three final revolute axes intersecting at one wrist center. State that this permits position-orientation decoupling.

Teaching note. Explain that the derivation is representative, not universal for every 6R architecture. The geometry of the selected robot must satisfy the partitioning assumption.

Image/figure note. Extend axes 4, 5, and 6 on an industrial arm until their common intersection is clearly visible.

Slide 3 — Desired tool pose and fixed tool offset (5 min)

On-slide content. Write $T_d = (R_d, \mathbf{p}_d)$ and define the fixed vector from wrist center to tool origin in the tool frame.

Teaching note. Distinguish the desired tool position from the wrist-center position. State the frame in which the offset vector is expressed.

Image/figure note. Draw the desired tool frame, the wrist center, and the offset vector as a rigid arrow attached to the tool frame.

Slide 4 — Compute the wrist center (5 min)

On-slide content. Derive

$$\mathbf{p}_w = \mathbf{p}_d - R_d \mathbf{p}_{w\to t}.$$

Teaching note. Explain the equation geometrically as subtracting the rotated tool offset from the desired tool origin. Avoid treating it as a formula to memorize.

Image/figure note. Use a vector triangle with the desired tool position, wrist center, and rotated offset color-matched to the equation.

Slide 5 — Solve joints 1–3 as a positioning subproblem (5 min)

On-slide content. State that the first three joints must place the wrist center at $\mathbf{p}_w$. Reuse the representative 3R analytical strategy.

Teaching note. Explicitly connect this step to the previous lecture. The arm solution may have shoulder and elbow alternatives, depending on the architecture.

Image/figure note. Fade out the wrist and tool so that the first three joints appear as a 3R arm reaching $\mathbf{p}_w$.

Slide 6 — Enumerate the arm branches — 5 min

On-slide content. Organize all valid (q_1, q_2, q_3) candidates in a branch tree and apply reachability and joint-limit tests.

Teaching note. Explain that branch counts are architecture-specific. Avoid presenting a universal maximum number without the assumptions that justify it.

Image/figure note. Show representative shoulder-left/right and elbow-up/down postures, linked to a branch tree.

Slide 7 — Residual orientation after solving the arm — 5 min

On-slide content. Compute

$$R_3^6 = (R_0^3)^T R_d.$$

Explain that this is the orientation still required from the spherical wrist.

Teaching note. Read the frame composition slowly. For each arm branch, R_0^3 differs, so the wrist orientation subproblem must be solved separately.

Image/figure note. Draw frames 0, 3, and 6 with arrows showing the composition from base to arm frame to desired tool orientation.

Slide 8 — Extract the three wrist angles — 5 min

On-slide content. Match R_3^6 to the chosen wrist-axis sequence and identify the matrix entries used to compute q_4, q_5, and q_6 with atan2.

Teaching note. Keep one rotation convention throughout. Show the matrix pattern and the extraction logic, but move lengthy trigonometric simplification to backup slides.

Image/figure note. Place the wrist-axis diagram beside the symbolic rotation matrix, with relevant entries highlighted.

Slide 9 — Wrist flip and nonflip solutions — 5 min

On-slide content. Show the two regular orientation decompositions associated with opposite signs of the middle wrist angle.

Teaching note. Explain how compensating changes in the first and third wrist angles preserve the same final orientation.

Image/figure note. Display two wrist postures with the same tool frame, using consistent axis colors.

Slide 10 — Complete 6R solution enumeration — 5 min

On-slide content. Combine each valid arm branch with every valid wrist branch, then filter limits and collisions and verify the full pose.

Teaching note. Present the structure as a Cartesian product followed by feasibility tests. A complete analytical set is a prerequisite for global branch comparison in planning.

Image/figure note. Draw the arm branch tree feeding into a wrist branch tree and ending in a candidate-solution table.

Slide 11 — Wrist singularity and nonuniqueness — 5 min

On-slide content. Show the configuration in which two wrist axes align and explain that q_4 and q_6 are no longer separately determined although their combined effect may remain defined.

Teaching note. Distinguish loss of a coordinate decomposition from loss of the desired orientation itself. Connect this to rank loss studied in Session 3.

Image/figure note. Compare a regular wrist with a singular aligned-axis wrist and draw the coincident axes prominently.

Slide 12 — From discrete 6R branches to 7R redundancy — 5 min

On-slide content. Explain that a generic 7R manipulator has one more joint variable than needed for a six-dimensional task pose, so the IK solutions form continuous families.

Teaching note. Distinguish discrete branch multiplicity from continuous redundancy. A 7R solver must select both a branch and a value of the redundancy parameter.

Image/figure note. Show discrete dots for 6R solutions and several continuous curves for 7R solutions.

Slide 13 — Shoulder–wrist geometry and the elbow circle — 5 min

On-slide content. Define the shoulder, elbow, and wrist points. Explain that fixing shoulder and wrist positions constrains the elbow to the intersection of two spheres, generically a circle.

Teaching note. Use this geometry to motivate the arm-angle parameter instead of introducing it algebraically.

Image/figure note. Draw two spheres centered at shoulder and wrist and highlight their intersection circle containing possible elbow positions.

Slide 14 — Arm angle or swivel angle ψ — 5 min

On-slide content. Define ψ as the angle locating the elbow around the shoulder–wrist axis relative to a chosen reference plane. State that $\mathbf{q} = \mathbf{q}^{(b)}(\psi)$ for branch b.

Teaching note. Define the reference-plane convention explicitly because several conventions exist in the literature. Explain that choosing ψ turns the redundant IK into a finite set of deterministic subproblems.

Image/figure note. Draw the shoulder–wrist axis, reference plane, elbow circle, elbow point, and the angle ψ.

Slide 15 — Feasible arm-angle intervals and planning relevance — 5 min

On-slide content. Show how joint limits, self-collision, external collision, and singularity margins remove parts of the nominal ψ range. Note that feasible intervals may be disconnected.

Teaching note. Finish by connecting the 7R family to motion planning: a path requires a continuous branch and arm-angle trajectory, not independent choices at each pose.

Image/figure note. Plot ψ on a circle or line with feasible green intervals, limit-violating red regions, collision regions, and singular neighborhoods.

5 Session 3, Lecture 1: Jacobian-Based Singularity Analysis of 3R Robots

Central question

How does local rank loss change attainable motion and partition the robot's joint space and workspace?

Slide 1 — A robot can reach a pose and still lose a motion direction 5 min

On-slide content. Compare a regular 3R posture with an aligned posture. Ask which end-effector velocity directions are possible in each case.

Teaching note. Introduce singularity through physical behavior before giving the rank definition. Encourage students to reason from joint-axis geometry.

Image/figure note. Show regular and singular configurations side by side, with attainable velocity arrows drawn at the end-effector.

Slide 2 — Differential kinematics 5 min

On-slide content. Starting from $\mathbf{x} = f(\mathbf{q})$, present $\dot{\mathbf{x}} = J(\mathbf{q})\dot{\mathbf{q}}$ and define the dimensions of every vector and matrix.

Teaching note. State exactly which task variables are used for the running 3R robot. Avoid using a six-dimensional twist if only position is being analysed.

Image/figure note. Draw a small joint-space displacement mapped to a small task-space displacement, with the Jacobian between them.

Slide 3 — Each Jacobian column has a physical meaning 5 min

On-slide content. State that column i is the end-effector velocity produced by unit speed at joint i with all other joints fixed.

Teaching note. Connect this directly to the screw-axis interpretation from PoE. Relative instantaneous motions add linearly even though finite rigid transformations do not commute.

Image/figure note. Show three copies of the 3R robot, each with one active joint and the resulting end-effector velocity arrow.

Slide 4 — Geometric derivation of the 3R Jacobian 5 min

On-slide content. For a revolute joint, use $\mathbf{z}_i \times (\mathbf{p} - \mathbf{o}_i)$ for the linear-velocity contribution. Assemble the three columns.

Teaching note. Define every origin and axis vector on the figure. Derive at least one column live and reveal the others progressively.

Image/figure note. Draw joint axes $\mathbf{z}_i$, origins $\mathbf{o}_i$, the end-effector point $\mathbf{p}$, and the radius vectors used in the cross products.

Slide 5 — Verification by differentiating forward kinematics 5 min

On-slide content. Differentiate the position equations and show that the resulting matrix matches the geometric Jacobian.

Teaching note. The goal is not to repeat algebra but to confirm consistency. Mention that symbolic differentiation is useful for verification, while geometry often gives better intuition.

Image/figure note. Use two arrows labelled "geometric construction" and "differentiation" converging to the same Jacobian matrix.

Slide 6 — Image, kernel, and rank — 5 min

On-slide content. Define $\operatorname{im} J$ as attainable task velocities and $\ker J$ as joint-rate combinations producing zero task velocity.

Teaching note. Explain the physical interpretation in regular and singular configurations. A nontrivial kernel indicates internal instantaneous joint motion for the selected output map.

Image/figure note. Draw vector-space sketches of a full-dimensional image, a reduced image, and a nonzero kernel direction.

Slide 7 — Definition of a singular configuration — 5 min

On-slide content. State that $\mathbf{q}^*$ is singular when $\operatorname{rank} J(\mathbf{q}^*)$ is lower than the maximum rank attained by the map.

Teaching note. Avoid saying simply "the determinant is zero" before checking whether the Jacobian is square and whether the chosen task map is appropriate.

Image/figure note. Show a regular matrix with independent columns beside a singular matrix with visibly dependent columns.

Slide 8 — Determinant or minors for the running 3R robot — 5 min

On-slide content. Compute the determinant of the appropriate square Jacobian, or the relevant maximal minors, and factor the singularity equation.

Teaching note. Interpret each factor geometrically. The algebra should end in recognizable joint-axis or link alignments.

Image/figure note. Place each factor beside a drawing of the robot configuration in which that factor vanishes.

Slide 9 — Physical effects of singularity — 5 min

On-slide content. List lost task directions, nonunique joint velocities, large joint speeds near singularity, numerical sensitivity, and possible controller saturation.

Teaching note. Distinguish exact rank loss from near-singular poor conditioning. Explain that practical problems appear before the exact singular set is reached.

Image/figure note. Use five simple icons or small robot sketches, one for each consequence.

Slide 10 — Singular values and velocity ellipsoids — 5 min

On-slide content. Explain that the unit joint-rate ball maps to a task-velocity ellipsoid whose principal semi-axes are determined by singular values.

Teaching note. Emphasize that $\sigma_{\min} \to 0$ corresponds to the ellipsoid flattening and one task direction becoming difficult or impossible.

Image/figure note. Animate or show a sequence of ellipsoids changing from well-rounded to flat as the robot approaches singularity.

Slide 11 — Conditioning and manipulability metrics 5 min

On-slide content. Introduce $\kappa(J) = \sigma_{\max}/\sigma_{\min}$ and, when dimensionally appropriate, $w = \sqrt{\det(JJ^T)}$.

Teaching note. Discuss scaling and units. Metrics that combine translational and rotational rows require care, and a scalar metric cannot describe every directional weakness.

Image/figure note. Plot $\sigma_{\min}$, condition number, and manipulability along one robot path.

Slide 12 — Caution before using J^TJ and JJ^T 5 min

On-slide content. Show $J \in \mathbb{R}^{m\times n}$, $J^TJ \in \mathbb{R}^{n\times n}$, and $JJ^T \in \mathbb{R}^{m\times m}$. Explain rank limits, dimensions, and the difference between joint and task redundancy.

Teaching note. State explicitly that the two products are not interchangeable. If $n > m$, J^TJ is necessarily singular even when J has full row rank; if $m > n$, JJ^T is necessarily singular even when J has full column rank.

Image/figure note. Use a dimension diagram with rectangular matrices and rank annotations. Add a warning symbol next to unjustified determinant tests.

Slide 13 — Singularity curves in joint space 5 min

On-slide content. Solve the singularity equation over an informative joint-space slice and respect the periodic identification of revolute coordinates.

Teaching note. Explain why singular sets typically form curves or surfaces rather than isolated points. Add joint limits only after showing the ideal periodic structure.

Image/figure note. Plot singular curves in (q_2, q_3), duplicate boundaries to show periodicity, and shade the allowed joint-limit region.

Slide 14 — Map the singularity set into workspace 5 min

On-slide content. Apply forward kinematics to singular configurations and show workspace boundaries and internal singular curves. Color regular regions by the number of IK solutions.

Teaching note. Explain that several joint-space points may map to the same workspace point. Crossing a singular image can change the number of regular inverse solutions.

Image/figure note. Use side-by-side joint-space and workspace plots connected by arrows and consistent colors.

Slide 15 — What the Jacobian tells us, and what it does not 5 min

On-slide content. Summarize: the Jacobian gives local velocity information and local conditioning. It does not by itself establish global IK branch connectivity, cuspidality, collision feasibility, or the topology of a complete path.

Teaching note. Finish with a concept check classifying statements as local or global. Preview the next lecture on aspects, uniqueness domains, and cuspidal connectivity.

Image/figure note. Draw a local tangent picture inside a magnifying glass, surrounded by a larger joint-space map containing globally connected regions.

6 Session 3, Lecture 2: IKS Distribution, Aspects, and Cuspidal Properties

Central question

When can distinct inverse-kinematic solutions be connected without crossing a singularity?

Slide 1 — Two postures, one pose: are they connected? 5 min

On-slide content. Show two distinct 3R configurations producing the same end-effector pose and ask whether a nonsingular path can connect them.

Teaching note. Do not answer immediately. Use the question to distinguish solution multiplicity from global connectivity.

Image/figure note. Display the two robot postures and a question-mark path between them. Keep the end-effector pose identical.

Slide 2 — The preimage of a workspace pose 5 min

On-slide content. Define $f^{-1}(\mathbf{x}_d)$ as the set of configurations producing the desired pose and place those configurations on a joint-space map.

Teaching note. For a nonredundant regular 3R robot, the preimage is generically finite. Explain that the positions of these points relative to singular curves determine connectivity questions.

Image/figure note. Connect one workspace target to several marked points in the (q_2, q_3) joint-space slice.

Slide 3 — Aspects as connected regular regions 5 min

On-slide content. Define an aspect as a maximal connected component of the joint space after removing singular configurations for the selected kinematic map.

Teaching note. Mention that joint limits and collision constraints can further divide the practically usable region even when the ideal aspect is connected.

Image/figure note. Shade the connected components separated by singular curves using distinct light colors.

Slide 4 — Noncuspidal robot behavior 5 min

On-slide content. State that, in a noncuspidal robot, distinct IKS of the same pose cannot be connected through a nonsingular path.

Teaching note. Explain the usual picture in which the solutions lie in different aspects. Clarify that this is a global property, not a local Jacobian test at one configuration.

Image/figure note. Show two IK points separated by a singular curve, with every candidate connecting path crossing the curve.

Slide 5 — Cuspidal robot behavior 5 min

On-slide content. State that a robot is cuspidal when at least two distinct IKS of the same pose can be connected by a nonsingular configuration-space path.

Teaching note. Emphasize that the end-effector need not remain fixed along the path; it returns to the same pose at the end. Avoid defining cuspidality only by the visual presence of a cusp-shaped curve.

Image/figure note. Show two IK points in the same connected regular region joined by a path bending around the singular set.

Slide 6 — Why elbow and shoulder labels are not always global 5 min

On-slide content. Explain that labels such as elbow-up and elbow-down describe local geometry but may change continuously along a nonsingular path in a cuspidal robot.

Teaching note. Use a sequence of configurations to show that a posture label can cease to be a reliable connected-component label.

Image/figure note. Create a storyboard of four to six robot postures along a continuous nonsingular posture-changing motion.

Slide 7 — Cusp points and solution coalescence 5 min

On-slide content. Introduce a cusp point as a special workspace point where three inverse solutions coalesce in the relevant reduced representation.

Teaching note. Give the geometric intuition first. Detailed algebraic multiplicity conditions may be placed in backup slides or the exercise notes.

Image/figure note. Draw a singular workspace curve with a cusp and illustrate three nearby solution sheets approaching the same point.

Slide 8 — Looping around a cusp changes the IK branch 5 min

On-slide content. Show a closed path in workspace encircling a cusp and the corresponding continuous joint path that ends on a different IK solution of the original pose.

Teaching note. This is the central visual explanation of a nonsingular posture change. Keep branch colors consistent across workspace and joint plots.

Image/figure note. Use a workspace loop around the cusp beside plots of $q_1(s)$, $q_2(s)$, and $q_3(s)$ showing continuous evolution.

Slide 9 — The folded-map interpretation 5 min

On-slide content. Explain that the forward map can fold one connected regular region over workspace so that two points in the same region share one workspace image.

Teaching note. Use the folded-sheet analogy carefully and immediately relate it back to the robot joint space.

Image/figure note. Draw a folded surface above a plane, with two points on the surface projecting to the same point below.

Slide 10 — Characteristic surfaces 5 min

On-slide content. Introduce characteristic surfaces as preimages of aspect boundaries that partition an aspect into regions useful for distinguishing IK solutions.

Teaching note. Keep the treatment operational. Students should understand why aspects alone may not provide one-to-one forward kinematics in a cuspidal robot.

Image/figure note. Show one aspect containing internal characteristic curves or surfaces that divide it into smaller regions.

Slide 11 — Uniqueness domains — 5 min

On-slide content. Define a uniqueness domain as a region in which the forward kinematic map is one-to-one for the selected output variables.

Teaching note. Explain how several uniqueness domains can lie inside one aspect and why they are useful for branch tracking.

Image/figure note. Use a colored joint-space map in which one aspect is subdivided into several labelled uniqueness domains.

Slide 12 — Compare a noncuspidal and a cuspidal 3R design — 5 min

On-slide content. Present two nearby parameter sets and compare their singularity curves, workspace regions, cusp points, and connectivity of IKS.

Teaching note. Keep the robot architectures comparable so that students see how link dimensions or offsets alter global kinematic structure.

Image/figure note. Arrange four plots in a two-by-two comparison: joint space and workspace for each design.

Slide 13 — Computational branch continuation — 5 min

On-slide content. Explain how to sample a workspace path, compute all analytical IKS at each sample, and connect nearby solutions while monitoring singularity crossings.

Teaching note. State that nearest-neighbor association is a useful local heuristic but must be checked against periodic angles, branch merging, and possible cusp-induced permutations.

Image/figure note. Show several colored IK curves over path parameter and arrows connecting candidate solutions between adjacent samples.

Slide 14 — Consequences for robot design and planning — 5 min

On-slide content. List the opportunity of nonsingular posture change and the challenges of branch certification, initialization, and path planning.

Teaching note. Present cuspidality neither as purely good nor purely bad. It expands possible connectivity but invalidates simple fixed-posture assumptions.

Image/figure note. Use a balanced "opportunity versus challenge" graphic with a representative robot motion on each side.

Slide 15 — Common misconceptions and final synthesis — 5 min

On-slide content. Correct three statements: multiple IKS do not automatically imply cuspidality; a cusp-shaped plot alone is not a complete proof; and Jacobian rank at one point does not decide global connectivity.

Teaching note. End with the sentence: the planner must follow connected configurations, not names assigned to postures. Preview Session 4, where this information becomes planning data.

Image/figure note. Draw a configuration-space path feeding into a layered motion-planning graph.

7 Session 4, Lecture 1: Joint-Space and Workspace Planning — Numerical and Analytical

Central question

How do we convert a desired path into a continuous, constraint-respecting joint trajectory?

Slide 1 — Path in workspace or path in joint space? 5 min

On-slide content. Show the same start and goal connected by a straight Cartesian path and by a straight joint-space interpolation. Compare the resulting robot motions.

Teaching note. Ask which path better expresses the task and which path is easier to make continuous in joint coordinates. State that the two planning spaces solve different problems.

Image/figure note. Use a split slide: Cartesian straight line on the left and straight line in (q_1, q_2) with its curved workspace trace on the right.

Slide 2 — Formal planning problem 5 min

On-slide content. Find a continuous path $\mathbf{q}(s)$, $s \in [0, 1]$, connecting start and goal while satisfying joint limits, collision constraints, kinematic feasibility, and possibly a prescribed task-space path.

Teaching note. Distinguish a geometric path from its later time parameterization. Explain that velocity and acceleration limits can be checked after a feasible geometric path has been selected.

Image/figure note. Draw a configuration-space path surrounded by icons for limits, collision, singularity margin, and task error.

Slide 3 — Joint-space interpolation 5 min

On-slide content. Present $\mathbf{q}(s) = (1 - s)\mathbf{q}_A + s\mathbf{q}_B$ with explicit angle wrapping for revolute joints.

Teaching note. State the advantages: joint continuity is direct and limits are easy to evaluate. State the limitation: the end-effector path is generally not a straight line and may enter obstacles.

Image/figure note. Show the straight joint-space segment and several corresponding robot snapshots in workspace.

Slide 4 — Configuration-space collision checking 5 min

On-slide content. Explain that collision-free endpoints do not guarantee a collision-free edge. A local path must be sampled or checked continuously using robot geometry.

Teaching note. Include self-collision and environmental collision. Explain why simplified collision meshes are often preferable to detailed visual meshes.

Image/figure note. Show three robot snapshots along one edge, with only the middle configuration colliding with an obstacle.

Slide 5 — Sampling-based planning in joint space 5 min

On-slide content. Introduce the abstraction: nodes are feasible configurations, edges are valid local motions, and search connects start to goal through the free configuration space.

Teaching note. Keep RRT and PRM details brief. The purpose is to emphasize that these methods explore connectivity in configuration space but do not automatically follow an exact Cartesian path.

Image/figure note. Draw configuration-space obstacles, random samples, graph edges, and one highlighted solution path.

Slide 6 — A prescribed Cartesian path — 5 min

On-slide content. Represent the desired path by ordered poses $\mathcal{P} = \{T_0, T_1, \ldots, T_N\}$ and state that a feasible configuration must be selected at every sample.

Teaching note. Explain that pointwise reachability is necessary but not sufficient. The selected configurations must form a continuous and collision-free sequence.

Image/figure note. Draw a Cartesian curve with ordered tool frames and a vertical column of IK candidates below each frame.

Slide 7 — Numerical IK continuation — 5 min

On-slide content. At sample $k+1$, initialize the numerical IK solver with the solution at sample k. Show the iterative correction based on the local Jacobian.

Teaching note. Present this as a common and effective local method. Explain that its behavior depends on the initial guess, step size, damping, and stopping criteria.

Image/figure note. Use a flowchart: next path pose → error → Jacobian update → new configuration.

Slide 8 — Damped least-squares update — 5 min

On-slide content. Present one dimensionally consistent DLS form, for example

$$\Delta \mathbf{q} = J^T (J J^T + \lambda^2 I)^{-1} \Delta \mathbf{x}.$$

Teaching note. Explain damping as a compromise between accuracy and bounded updates near poor conditioning. Remind students to check matrix dimensions and not to treat DLS as a global planner.

Image/figure note. Highlight the damping term inside a block diagram of the inverse differential-kinematics calculation.

Slide 9 — Failure mode: initialization and local convergence — 5 min

On-slide content. Show several initial guesses converging to different IK branches or failing for the same target.

Teaching note. Explain basins of attraction and why the previous solution usually helps but does not provide a global guarantee.

Image/figure note. Plot four numerical iterations from different seeds, ending at different analytical IK solutions.

Slide 10 — Failure mode: branch jumping and discontinuity — 5 min

On-slide content. Show a Cartesian path with small task error but a sudden jump in one or more joint variables.

Teaching note. State that independently good pose solutions can form a physically impossible trajectory. Define a wrapped joint-distance continuity test.

Image/figure note. Use synchronized Cartesian-error and joint-angle plots, with the branch jump marked in red.

Slide 11 — Failure mode: singularity and hidden dead ends — 5 min

On-slide content. Combine two examples: joint updates grow near a singularity, and a greedy branch later reaches a joint limit although another branch remains feasible.

Teaching note. Explain that local damping cannot reveal a globally better branch. This is the motivation for preserving all analytical alternatives.

Image/figure note. Show two branch curves over path index: one approaches a singular/limit boundary and terminates, while another continues.

Slide 12 — Analytical IK enumeration at every path sample — 5 min

On-slide content. Define $\mathcal{Q}_k = \{\mathbf{q}_k^{(1)}, \ldots, \mathbf{q}_k^{(m_k)}\}$, then filter joint limits, collision, and minimum singularity margin.

Teaching note. Explain that enumeration delays the branch decision and preserves alternatives for a global sequence choice.

Image/figure note. Draw each path sample above a vertical layer of candidate configurations, with invalid nodes crossed out.

Slide 13 — Layered IK graph — 5 min

On-slide content. Define nodes as valid IKS and edges as feasible transitions between consecutive path samples. Require bounded wrapped joint displacement and collision-free interpolation.

Teaching note. Make the graph construction explicit. A node is not merely an IK branch name; it contains a numerical joint vector and feasibility metadata.

Image/figure note. Draw a layered directed acyclic graph with feasible edges in gray and infeasible transitions omitted.

Slide 14 — Costs and graph search — 5 min

On-slide content. Define transition costs for joint displacement, distance from limits, singularity robustness, collision clearance, and optional branch preferences. Explain dynamic programming or shortest-path search.

Teaching note. Hard constraints must remove nodes or edges; soft costs only rank feasible alternatives. For a strictly layered acyclic graph, dynamic programming can replace general Dijkstra search.

Image/figure note. Highlight the minimum-cost path through the layered graph and show an edge-cost decomposition beside one edge.

Slide 15 — Numerical versus analytical planning — 5 min

On-slide content. Compare local numerical continuation and analytical layered planning using continuity, completeness over enumerated branches, architecture dependence, computation, and robustness near singularities.

Teaching note. Finish with a benchmark example and state the main result: solving IK at every point is not equivalent to planning a continuous joint-space trajectory. Preview cuspidal connectivity and redundancy parameterization.

Image/figure note. Use a compact dashboard with success, maximum joint jump, minimum $\sigma_{\min}$, joint-limit margin, and runtime for both methods.

8 Session 4, Lecture 2: Motion Planning for Cuspidal and Redundant Robots

Central question

How do we plan over true IK connectivity while selecting a continuous redundancy parameter?

Slide 1 — Why nearest IK is not enough 5 min

On-slide content. Show a cuspidal 3R robot changing posture nonsingularly and a 7R robot moving its elbow around an obstacle while maintaining the same tool pose.

Teaching note. State that the final lecture adds two planning complications: global connectivity can violate simple posture labels, and redundant IK contains a continuous family of choices.

Image/figure note. Use a two-panel sequence: a cusp-mediated posture change and a 7R elbow-swivel motion.

Slide 2 — Kinematics-grounded planning data 5 min

On-slide content. List the data available from earlier sessions: robot model, complete IK branches, singularity sets, aspects or connectivity labels, joint limits, and collision geometry.

Teaching note. Explain that a kinematics-grounded planner exposes this structure instead of calling an IK solver as an opaque subroutine.

Image/figure note. Draw all information sources feeding into one planner block and then into simulation validation.

Slide 3 — Planning for a noncuspidal robot 5 min

On-slide content. Explain that distinct IKS of the same pose are separated by singularities, so remaining inside one aspect preserves a regular branch.

Teaching note. State when fixed branch labels are a reasonable simplification. Also note that limits and collisions may still break a branch within an aspect.

Image/figure note. Show separated aspects with one continuous solution curve inside each.

Slide 4 — Why the same rule fails for a cuspidal robot 5 min

On-slide content. Show two distinct IKS in the same regular connected component and a nonsingular path between them.

Teaching note. Contrast two errors: forbidding every posture change rejects feasible paths, while jumping directly between IK list entries creates discontinuity.

Image/figure note. Draw one aspect with two IK points, a valid curved connection, and an invalid straight jump marked with a red cross.

Slide 5 — Represent true branch connectivity 5 min

On-slide content. Attach aspect, uniqueness-domain, or sampled connected-component information to analytical IK candidates and edges.

Teaching note. Explain that connectivity must refer to physical continuous paths, not only equality of heuristic labels such as elbow-up or wrist-flip.

Image/figure note. Color graph nodes by connected component and use different boundary styles for uniqueness domains.

Slide 6 — Plan a nonsingular posture change 5 min

On-slide content. Construct a workspace loop around a cusp or a joint-space path inside one regular component. The start and goal have the same pose but different configurations.

Teaching note. Emphasize that the end-effector usually moves away from the target pose during the posture change and returns to it at the end.

Image/figure note. Show the closed workspace loop, the open joint-space path, and several intermediate robot postures.

Slide 7 — Redundant IK as branch families 5 min

On-slide content. For a 7R robot, write $\mathbf{q} = \mathbf{q}^{(b)}(\psi)$, where b is a discrete branch and ψ is the arm angle.

Teaching note. Explain that a planner must select both the discrete family and a continuous parameter trajectory. Independent per-pose optimization can produce discontinuities.

Image/figure note. Plot several colored solution curves over a horizontal ψ axis, one curve per discrete branch.

Slide 8 — Feasible arm-angle set at one path sample 5 min

On-slide content. Define Ψ_k as the set of arm angles satisfying joint limits, self-collision, environmental collision, and singularity-margin constraints at pose T_k.

Teaching note. Explain that Ψ_k may consist of several disconnected intervals. A single locally optimal ψ does not describe the complete feasible set.

Image/figure note. Use a circular or linear ψ plot with feasible green intervals and forbidden regions labelled by their cause.

Slide 9 — Feasible arm-angle map along a trajectory 5 min

On-slide content. Stack Ψ_k over path index to obtain a two-dimensional map of feasible corridors, bottlenecks, and disconnected regions.

Teaching note. Explain that this map is the redundant analogue of preserving all discrete IK nodes in a layered graph.

Image/figure note. Draw a heat map with path index vertically, ψ horizontally, and feasible regions highlighted.

Slide 10 — Continuity of the redundancy parameter 5 min

On-slide content. Require a continuous or bounded-step sequence $\psi_0, \ldots, \psi_N$, with correct angular wrapping.

Teaching note. Show why choosing the best ψ independently at every sample can jump between disconnected intervals or cause large joint motion.

Image/figure note. Overlay a smooth feasible $\psi(k)$ curve and an independently optimized discontinuous curve on the feasibility map.

Slide 11 — Layered graph over branch and arm angle 5 min

On-slide content. Discretize each feasible interval, create nodes (b, ψ), and connect nearby states across consecutive path samples only when the interpolated robot motion is feasible.

Teaching note. Explain that this construction generalizes the 6R layered IK graph from discrete nodes to sampled branch families.

Image/figure note. Draw several nodes along each branch curve in every layer, with one highlighted globally consistent path.

Slide 12 — Costs for redundant planning 5 min

On-slide content. Combine joint smoothness, distance from limits, collision clearance, singularity robustness, preferred posture, and arm-angle variation. Separate hard constraints from soft costs.

Teaching note. Normalize quantities before weighting them. Explain that a large weight cannot safely replace removal of an actually colliding or invalid state.

Image/figure note. Use a cost stack beside two candidate nodes, showing why the globally chosen node is preferable.

Slide 13 — Null-space redundancy resolution 5 min

On-slide content. Introduce

$$\dot{\mathbf{q}} = J^{\#}\mathbf{v} + (I - J^{\#}J)\mathbf{z}$$

as a local method for performing secondary motion without changing the primary task instantaneously.

Teaching note. Explain the primary term, null-space projector, and secondary objective. State clearly that the method is local and depends on the chosen pseudoinverse and gradient.

Image/figure note. Show the end-effector following a prescribed velocity while the elbow moves in a null-space direction.

Slide 14 — Limits of local null-space optimization 5 min

On-slide content. Show that local gradients can become trapped, approach feasibility bottlenecks, or fail to move between disconnected feasible ψ intervals.

Teaching note. Present analytical parameterization and null-space control as complementary. A hybrid method can use global arm-angle planning to provide a corridor and local control for execution.

Image/figure note. Draw a local gradient path trapped in one feasible pocket while a global corridor exists in another part of the map.

Slide 15 — Integrated pipeline and final synthesis 5 min

On-slide content. Give the complete pipeline: sample the task path, enumerate branch families, compute feasible sets, build a connectivity graph, search, refine, time-parameterize, and verify in simulation. End with the sequence: read, solve, analyse, plan.

Teaching note. Reiterate the caution that the Jacobian is a local velocity map, not a silver bullet for global IK multiplicity, branch connectivity, collision, joint limits, or closed-loop constraints. State the expected final-project outputs and evaluation metrics.

Image/figure note. Use a full-width end-to-end flowchart followed by a dashboard containing task error, joint continuity, minimum limit margin, collision clearance, minimum singular value, and runtime.

Final deck-production checklist

Before finalizing each deck, verify the following:

- Every symbol is defined before it is used, and the twist/Jacobian coordinate ordering is unchanged across lectures.
- Every important equation has a geometric interpretation, figure, or simulation example.
- Slide titles state a conclusion or question rather than only naming a topic.
- Complete algebraic expansions, alternative conventions, and extra cases are placed in backup slides.
- Figures use readable labels, consistent colors, and captions that identify what students should notice.
- Each lecture contains at least one conceptual check and ends with three to five explicit takeaways.
- The 45-minute exercise begins from the final lecture concept and uses prepared starter material rather than requiring infrastructure to be built from scratch.